Diagonalization

Geometric Algorithms

CAS CS 132

Objectives

- Finish our discussion on the characteristic polynomial
- Motivate diagonalization via linear dynamical systems and changes of coordinate systems
- Describe how to diagonalize a matrix

Keywords

multiplicity

similar matrices

diagonalizable matrices

change of basis

eigenbasis

Recap: Characteristic Polynomial

Recall: Determinants and Invertibility

$\det(A)$ is an value associate with the matrix A

Thm. A matrix is invertible if and only if $\det(A) \neq 0$

So by the Invertible Matrix Theorem:

$\det(A - \lambda I) = 0$
polynomial in λ

$\equiv$ $(A - \lambda I)\mathbf{x} = \mathbf{0}$ **has nontrivial solutions**

$\equiv$ λ is an eigenvalue of A

How To: Finding Eigenvalues

Q. Determine the eigenvalues of A

A. Find *roots* of the *characteristic poly* of A:

$$\det(A - \lambda I)$$

Computationally, we can also use:

numpy.linalg.eig(A)

Example

$$A = \begin{bmatrix} 1 & -1 \\ 4 & -3 \end{bmatrix}$$

An Observation: Multiplicity

$$\lambda^1(\lambda-1)^2(\lambda-4)^1$$ multiplicities

In the examples so far, we've seen a number appear as a root multiple times

This is called the **algebraic multiplicity** of the root

Is the algebraic multiplicity meaningful?

Multiplicity and Dimension

Thm. The dimension of the eigenspace of A for the eigenvalue λ is *at most* the algebraic multiplicity of λ in $\det(A - \lambda I)$

The algebraic multiplicity is an upper bound on "how large" the eigenspace is

Example Practice Problems

Let A *be a* 5×5 *matrix with characteristic polynomial* $(x-1)^3(x-3)(x+5)$

Q1» *What is* $\text{rank}(A)$*?*

Q2» *What is the minimum possible rank of* $A - I$*?*

Q3. Find the eigenvalues and an eigenbasis for:

$$\begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

Example

Let A *be a* 5×5 *matrix with characteristic polynomial* $(x-1)^3(x-3)(x+5)$

» *What is* $\text{rank}(A)$*?*

» *What is the minimum possible rank of* $A - I$*?*

Answer

$$\begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

Answer

$$\begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

Motivating Diagonalization via Linear Dynamical Systems

Recall: Eigenbasis

Def. An eigenbasis of H for $A \in \mathbb{R}^{n \times n}$ is a basis of H made up of eigenvectors of A

We will be almost exclusively interested of eigenbases of $\mathbb{R}^n$ when $A \in \mathbb{R}^{n \times n}$

When can we describe any vector in $\mathbb{R}^n$ as a unique linear combination of eigenvectors of A?

Recall: Linear Dynamical Systems

$$\mathbf{v}_1 = A\mathbf{v}_0$$
$$\mathbf{v}_2 = A\mathbf{v}_1 = A^2\mathbf{v}_0$$
$$\mathbf{v}_3 = A\mathbf{v}_2 = A^3\mathbf{v}_0$$
$$\mathbf{v}_4 = A\mathbf{v}_3 = A^4\mathbf{v}_0$$
$$\vdots$$

multiplying by A changes the state

A **linear dynamical system** describes a sequence of **state vectors** starting at $\mathbf{v}_0$

demo

Also, check out ILA demos (near bottom of this page): https://textbooks.math.gatech.edu/ila/diagonalization.html

Eigenbases and Closed-Form solutions

Given $\mathbf{v}_k = A\mathbf{v}_{k-1} = A^k\mathbf{v}_0$, if

eigenvectors of A

$$\mathbf{v}_0 = \alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 + \alpha_3\mathbf{b}_3$$

then

eigenvalues of A

$$A^k\mathbf{v}_0 = \alpha_1\lambda_1^k\mathbf{b}_1 + \alpha_2\lambda_2^k\mathbf{b}_2 + \alpha_3\lambda_3^k\mathbf{b}_3$$

closed-form solution

Application: Eigenbases and Limiting Behavior

Thm. If A has an eigenbasis with eigenvalues

$$|\lambda_1| \geq |\lambda_2| \ldots \geq |\lambda_k|$$

then $\mathbf{v}_k \sim \lambda_1^k \mathbf{u}$ for some vector $\mathbf{u}$

The system grows exponentially in λ_1

The Takeaway

Given a basis $\mathscr{B}$ for $\mathbb{R}^n$, we only need to know how $A \in \mathbb{R}^n$ behaves on $\mathscr{B}$

Sometimes, A behaves simply on $\mathscr{B}$, as in the case of *eigenbases*

What we're really doing is changing our coordinate system to expose a behavior of A

Recap: Change of Basis

Recall: Bases define Coordinate Systems

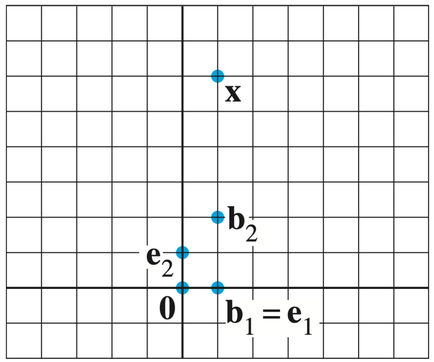


FIGURE 1 Standard graph paper.

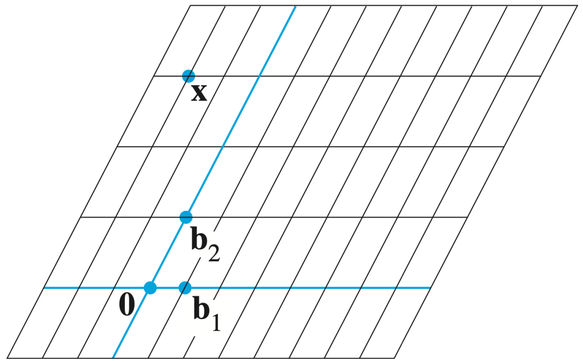


FIGURE 2 $\mathcal{B}$-graph paper.

Given a basis $\mathcal{B}$ of $\mathbb{R}^n$, there is **exactly one way** to write every vector as a linear combination of vectors in $\mathcal{B}$

Every basis provides a way to write down *coordinates* of a vector

$\mathcal{B}$ defines a "different grid for our graph paper"

Linear Algebra and its Applications (Lay, Lay, McDonald)

Recall: Coordinate Vectors

Let $\mathbf{v}$ be a vector in a $\mathbb{R}^n$ and let $\mathscr{B} = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$ be a basis of $\mathbb{R}^n$ where $\mathbf{v} = a_1\mathbf{b}_1 + a_2\mathbf{b}_2 + \ldots + a_n\mathbf{b}_n$

Df. The **coordinate vector of $\mathbf{v}$ relative to $\mathscr{B}$** is

$$[\mathbf{v}]_{\mathscr{B}} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Question

We know that if a $n \times n$ matrix $B = [\mathbf{b}_1\ \mathbf{b}_2\ \dots\ \mathbf{b}_n] \in \mathbb{R}^{n \times n}$ is invertible, then the columns of B form a basis $\mathscr{B}$ of $\mathbb{R}^n$

What is the matrix that implements the transformation

$$\mathbf{x} \mapsto [\mathbf{x}]_{\mathscr{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

where $\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \dots + c_n\mathbf{b}_n$*?*

Change of Basis Matrix

Thm. If $\mathscr{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n\}$ form a basis of $\mathbb{R}^n$, then

$$[\mathbf{x}]_{\mathscr{B}} = [\mathbf{b}_1 \ \mathbf{b}_2 \ \ldots \ \mathbf{b}_n]^{-1}\mathbf{x}$$

Matrix inverses perform changes of bases

How To: Change of Basis

Q. Given a basis $\mathscr{B} = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$ of $\mathbb{R}^n$, find the matrix which implements $\mathbf{x} \mapsto [\mathbf{x}]_{\mathscr{B}}$

A. Construct the matrix $[\mathbf{b}_1\ \mathbf{b}_2\ \ldots\ \mathbf{b}_n]^{-1}$

Example

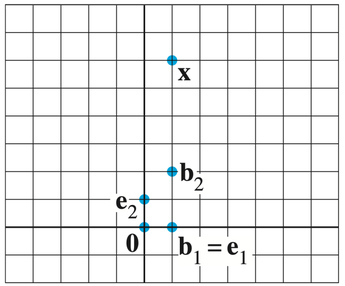


FIGURE 1 Standard graph paper.

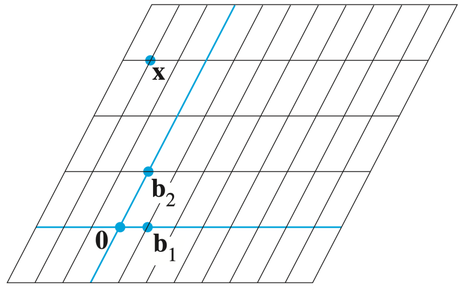


FIGURE 2 $\mathcal{B}$-graph paper.

Write the change-of-bases matrix for the basis $\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right)$

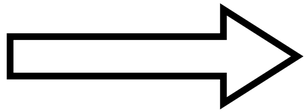
Diagonalization

Diagonal Matrices

ex. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -0.4 & 0 & 0 \\ 0 & 0 & 22 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Def. A matrix $A \in \mathbb{R}^{n \times n}$ is **diagonal** if

$$i \neq j \quad \text{implies } A_{ij} = 0$$



Only the diagonal entries can be nonzero

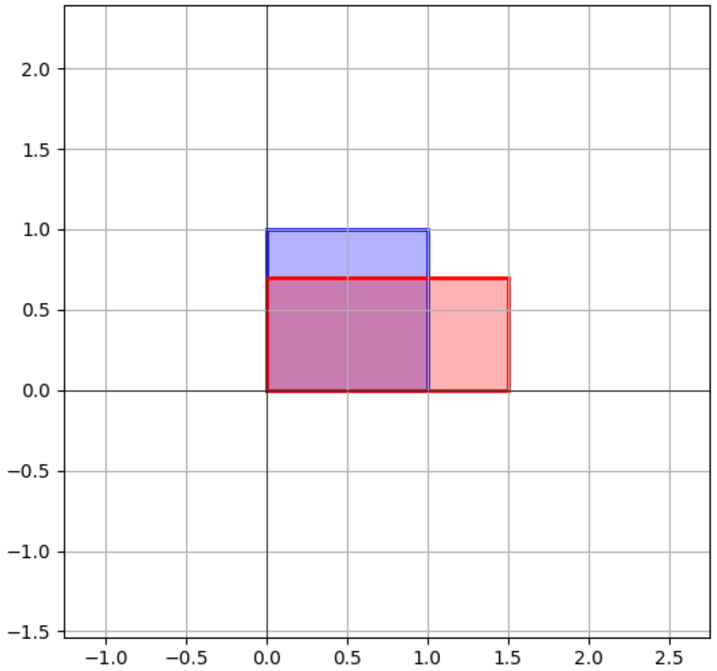
Diagonal matrices are (unequal) scaling matrices

Recall: Unequal Scaling

The scaling matrix *affects each component of a vector in a simple way*

The diagonal entries <u>scale</u> each corresponding entry

$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1.5x \\ 0.7y \end{bmatrix}$$



$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix}$$

When do matrices "behave" like unequal scaling matrices "up to" change of basis?

Scaling and Eigenvectors

The idea. Matrices behave like unequal scaling matrices on eigenvectors

$$\begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix} (x\mathbf{e}_1 + y\mathbf{e}_2) = x2\mathbf{e}_1 + y(-3)\mathbf{e}_2$$

$$A \begin{bmatrix} x \\ y \end{bmatrix}_{\mathscr{B}} = A(x\mathbf{b}_1 + y\mathbf{b}_2) = x\lambda_1\mathbf{b}_1 + y\lambda_2\mathbf{b}_2$$

Can we expose this behavior in terms of a matrix factorization?

Recall: Matrix Factorization

A **factorization** of a matrix A is an equation which expresses A as a product of one or more matrices, e.g.,

$$A = PBP^{-1}$$

Similar Matrices

$$A = PBP^{-1}$$

Def. A matrix A is **similar** to a matrix B if there is some invertible matrix P such that $A = PBP^{-1}$

A and B are the same up to a change of basis

Similar Matrices and Eigenvalues

Thm. Similar matrices have the *same eigenvalues*

Diagonalizable Matrices

Def. A matrix A is **diagonalizable** if it is similar to a diagonal matrix

There is an invertible matrix P and diagonal matrix D such that $A = PDP^{-1}$

Diagonalizable matrices are the same as scaling matrices up to a change of basis

Important: Not all Matrices are Diagonalizable

This is very different from the LU factorization

We will need to figure out which matrices are diagonalizable

Question: Is the zero matrix diagonalizable?

Application: Matrix Powers

only take the power of B

Thm. If $A = PBP^{-1}$, then $A^k = PB^kP^{-1}$

It may be easier to take the power of B (as in the case of diagonal matrices)

But how do we find the diagonalization...

Diagonalization and Eigenvectors

Suppose we have a diagonalization

$$A = PDP^{-1}$$

What do we know about it?

Columns of P are eigenvectors

$$A = [\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3] \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} [\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3]^{-1}$$

Columns of P are eigenvectors

$$A = \overbrace{[\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3]}^{\text{eigenbasis}} \overbrace{\begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}}^{\text{eigenvalues}} [\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3]^{-1}$$

In fact, the columns of P form an **eigenbasis** of $\mathbb{R}^n$ for A

And the entries of D are the **eigenvalues** associated to each eigenvector

A diagonalization exposes a lot of information about A

The Diagonalization Theorem

Thm. A matrix is diagonalizable if and only if it has an eigenbasis

We can use the same recipe to go in the other direction, given an eigenbasis, we can **build a diagonalization**

Diagonalizing a Matrix

High Level

$$A = PDP^{-1}$$

The columns of P form an *eigenbasis* for A

The diagonal of D are the eigenvalues for each column of P

The matrix P^{-1} is a change of basis to this eigenbasis of A

Step 1: Eigenvalues

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

Find all the eigenvalues of A

Find the roots of $\det(A - \lambda I)$

e.g. $\det(A - \lambda I) = -(\lambda - 1)(\lambda + 2)^2$

Upshot: nasty so we give it here

$\therefore$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 3 & 3 \\ -3 & -5-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{bmatrix}$$

$$\det = (1-\lambda)\left[(-5-\lambda)(1-\lambda) + 9\right]$$
$$-3\left[-3(1-\lambda) + 9\right]$$
$$+3\left[-9 + 3(5+\lambda)\right]$$

Step 2: Eigenvectors

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$\lambda_1 = 1$$

$$\lambda_2 = -2$$

Find **bases** of the corresponding eigenspaces

e.g.

$$\begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 3 & 3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & -3 & -3 \\ 0 & 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$v_1 = v_3$
$v_2 = -v_3$
v_3 free

$$\text{Nul}(A - I) = \text{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$$\vec{v} = v_3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$v_1 = -v_2 - v_3$
v_2 free
v_3 free

$$\begin{bmatrix} 3 & 3 & 3 \\ -3 & -3 & -3 \\ 3 & 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Nul}(A + 2I) = \text{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\vec{v} = v_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + v_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Step 3: Construct P

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$\lambda_1 = 1$$

$$\lambda_2 = -2$$

$$\text{Nul}(A - I) = \text{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$$\text{Nul}(A + 2I) = \text{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

If there are n eigenvectors from the previous step they form an **eigenbasis**

Build the matrix with these vectors as the columns

e.g.

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Step 5: Construct D

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$\lambda_1 = 1$$

$$\lambda_2 = -2$$

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Build the matrix with eigenvalues as diagonal entries

Note the order. It should be the same as the order of columns of P

e.g.

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Step 6: Invert P

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Find the inverse of P

$$\left[\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 & -1 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 2 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 & -1 & 0 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -1 & -1 & 0 \end{array}\right]$$

$$P^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

Putting it Together

$$\underset{A}{\begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}} = \underset{P}{\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}} \underset{D}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}} \underset{P^{-1}}{\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1}}$$

We know how to do every step, its a matter of putting it all together

How to: Diagonalizing a Matrix

Question. Find a diagonalization of $A \in \mathbb{R}^n$, or determine that A is not diagonalizable

Solution.

1. Find the eigenvalues of A, and bases for their eigenspaces. If these eigenvectors don't form a basis of $\mathbb{R}^n$, then A is **not diagonalizable**
2. Otherwise, build a matrix P whose columns are the eigenvectors of A
3. Then build a diagonal matrix D whose entries are the eigenvalues of A *in the same order*
4. Invert P
5. The diagonalization of A is PDP^{-1}

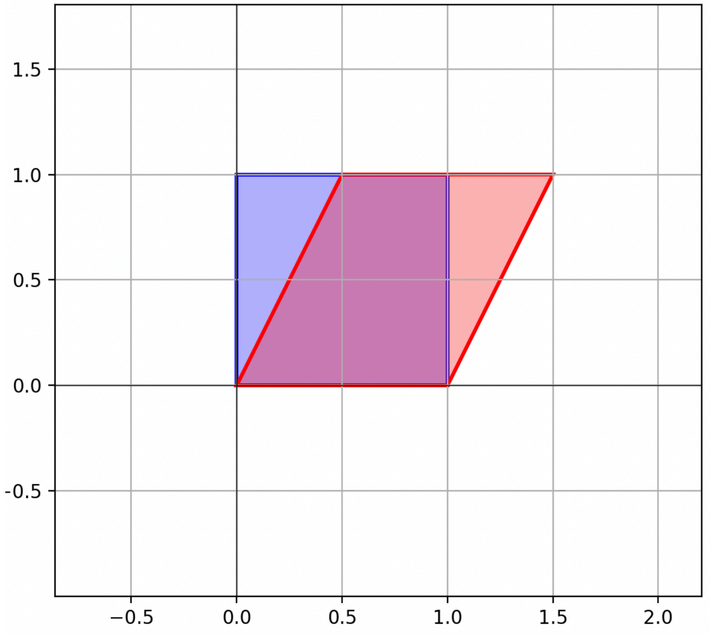
Example of Failure: Shearing

$$A = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$$

The shearing matrix has a single eigenvalue with an eigenspace of dimension 1

We can't build an eigenbasis of $\mathbb{R}^2$ for A

In other words, A is not diagonalizable



Important case: Distinct Eigenvalues

$$\begin{bmatrix} 1 & -3 & 4 & 2 \\ 0 & -2 & 3 & -1 \\ 0 & 0 & 10 & 5 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

Thm. If a matrix in $\mathbb{R}^{n \times n}$ has n distinct eigenvalues, then it is diagonalizable

This is because eigenvectors with distinct eigenvalues are *linearly independent*

Example

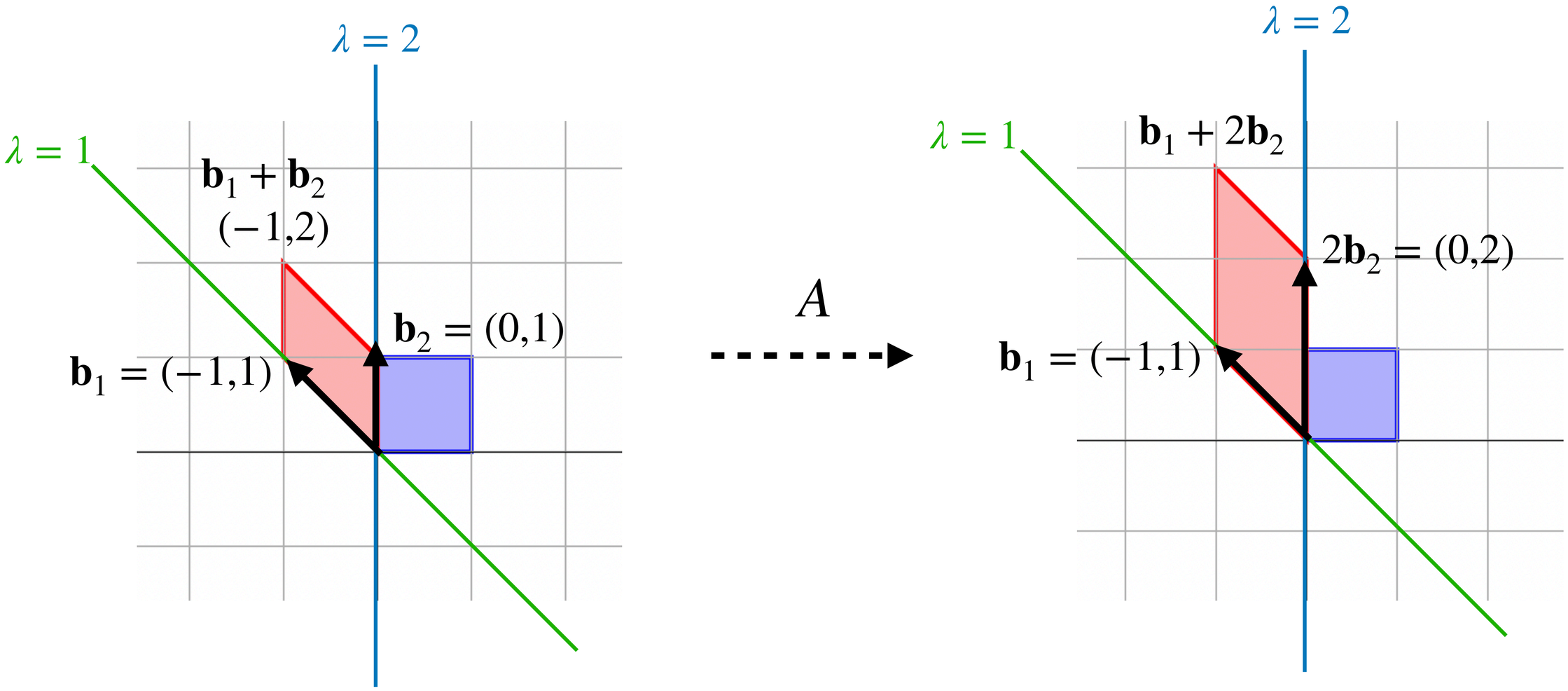
$$\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$$

Find a diagonalization of the above matrix

The Picture

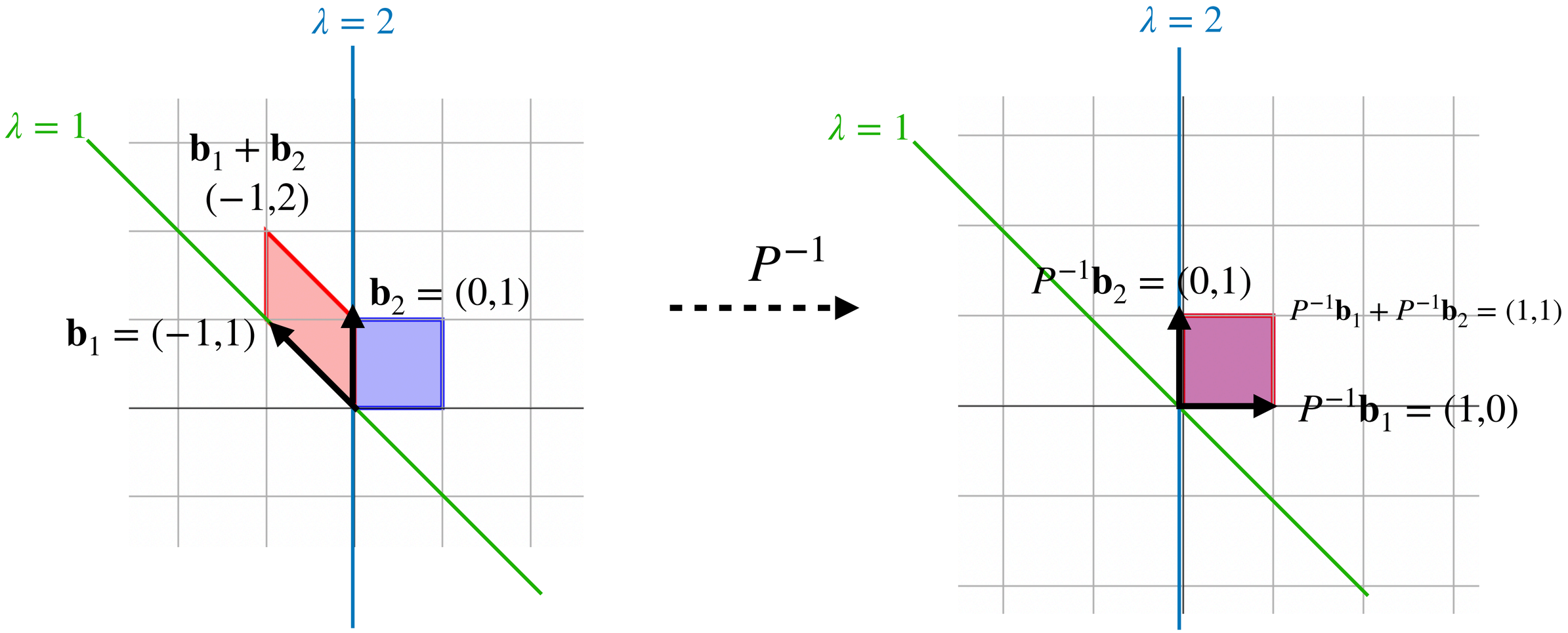
Example (Geometric)

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$$



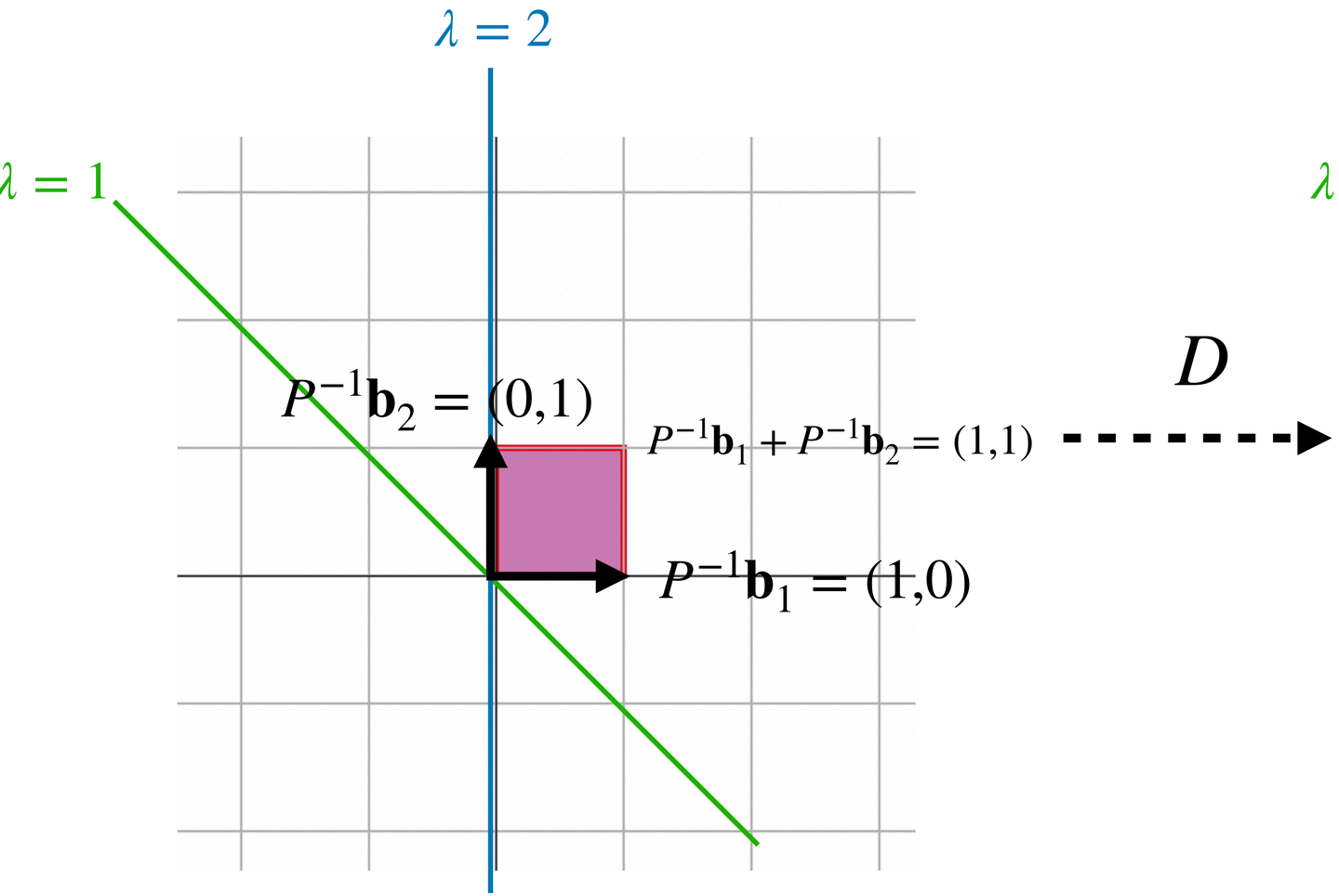
Example (Geometric)

$$P^{-1} = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}^{-1}$$

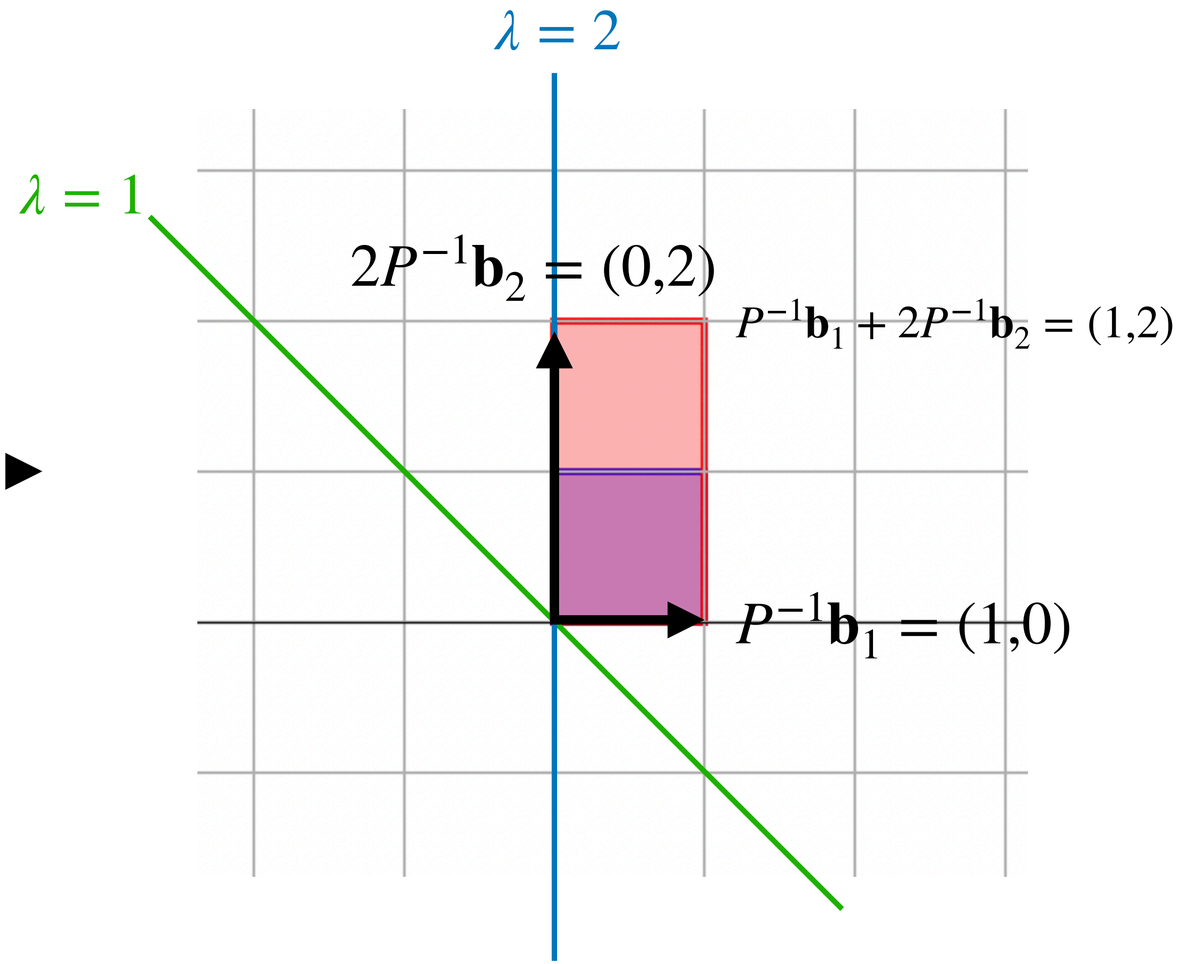


Example (Geometric)

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

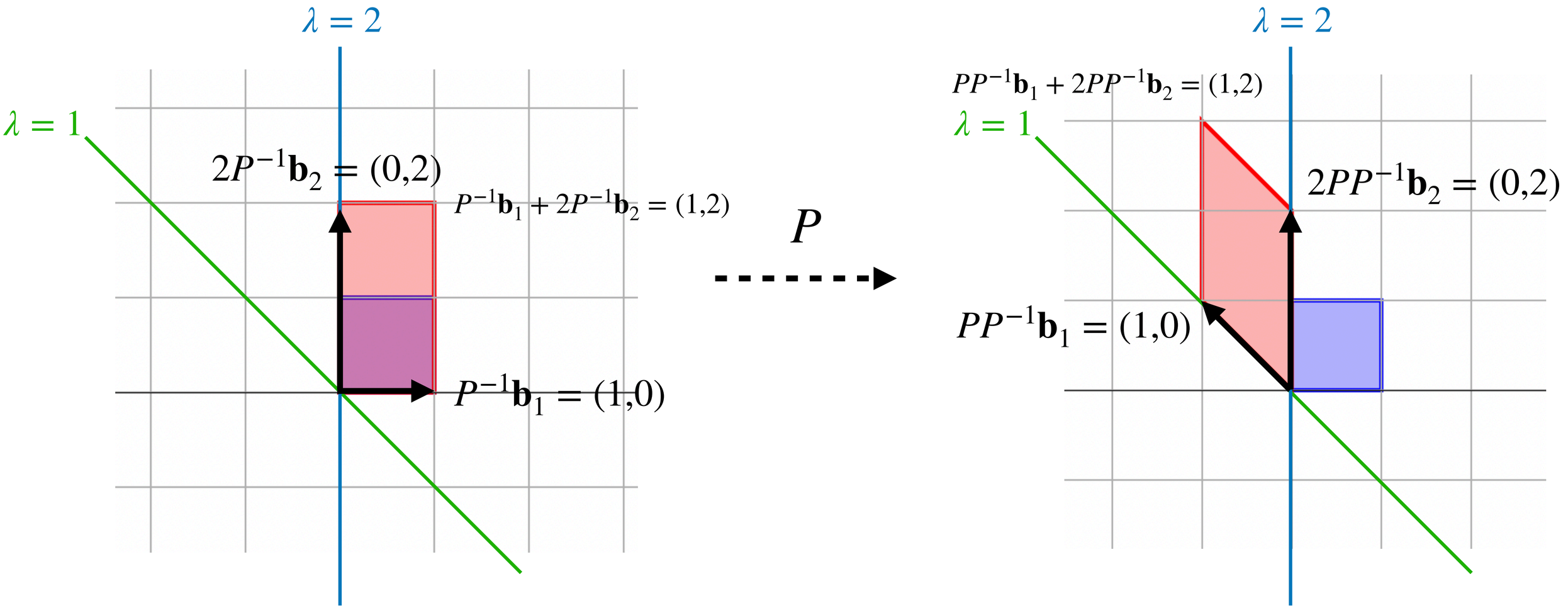


D

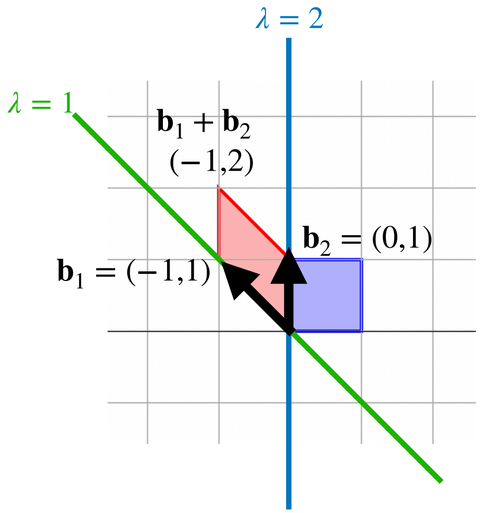


Example (Geometric)

$$P = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}$$

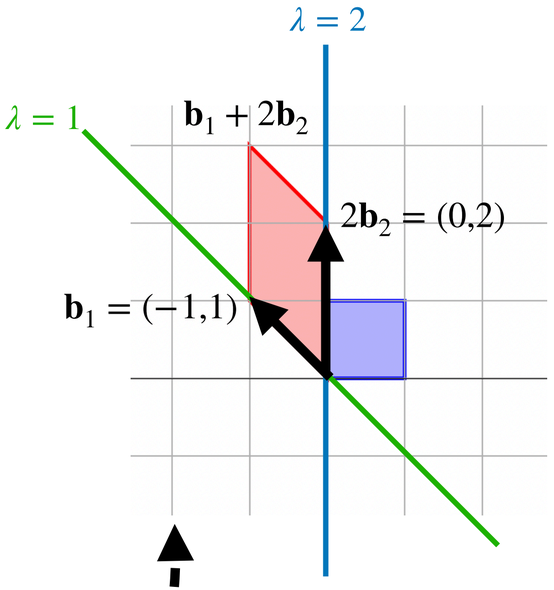


Example (Geometric)

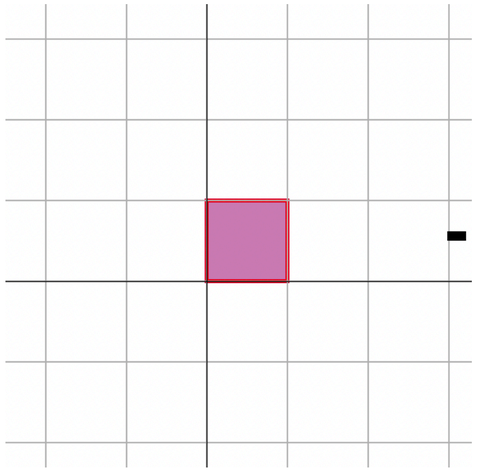


$$A = PDP^{-1}$$

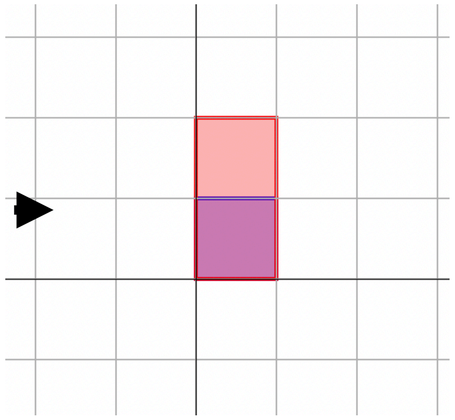
$$\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}^{-1}$$



P^{-1}



D



P